**QPCODE** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) PRACTICAL EXAMINATION OCTOBER 2025

## THIRD SEMESTER

## DSE – MG3DSEPHY201 – COMPUTATIONAL PHYSICS- C++ PROGRAMMING

(2024 ADMISSION ONWARDS)

**Duration: 2 HOURS** **Maximum Marks: 35**

**Record: 5 Marks**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

| | | |
|---|---|---|
| 1. | Write a C++ program to solve a quadratic equation of the form $ax^2 + bx + c = 0$.. | A/CO4,CO5, CO6,CO7 |
| 2. | Write a C++ program to convert a decimal number into its binary equivalent.. | A/CO4,CO5, CO6,CO7 |
| 3. | Write a C++ program to sort an array of 10 integers in ascending and descending order. | A/CO4,CO5, CO6,CO7 |
| 4. | Write a C++ program to add two matrices of order m x n.. | A/CO4,CO5, CO6,CO7 |
| 5. | Write a C++ program to generate the first n prime numbers. | A/CO4,CO5, CO6,CO7 |
| 6. | Write a C++ program to compute **sin(x)/cos(x)/exp(x)** using series expansion. Compare with values from <cmath> (math.h) and tabulate the error with increasing number of terms. | An, A/CO4,CO5, CO6,CO7 |
| 7. | Write a C++ program to perform matrix multiplication. | A/CO4,CO5, CO6,CO7 |

| | | | |
|---|---|---|---|
| 8. | Create a user-defined data type **Complex** and implement addition, subtraction, and multiplication using function overloading and operator overloading. | C,A/CO4,CO5,CO6,CO7 |
| 9. | Define a user-defined data type Vector to represent 1D vectors and implement vector addition, subtraction, and dot product using operator and function overloading. | C,A/CO4,CO5,CO6,CO7 |
| 10. | Write a C++ program to find the determinant of a square matrix (2x2 or 3x3). | A/CO4,CO5,CO6,CO7 |